MACHINE LEARNING · HEALTHCARE

ECG-Based Cardiac Arrhythmia Detection Using Machine Learning

An ML-based approach to ECG beat classification using signal processing and engineered features

Presented by

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Problem Statement and Motivation

The problem

ECG signals carry important information about the electrical activity of the heart, but different heartbeat patterns can be hard to distinguish. Automated classification becomes challenging when abnormal beat classes are under-represented.

Why does it matter?

01

Complex ECG patterns

Beat morphology and timing vary between heartbeat categories.

02

Class imbalance

Normal beats substantially outnumber some abnormal beat types.

03

Reliable evaluation

A model must be tested on unseen ECG records, not only familiar data.

An automated tool can assist, not replace, the clinician: it processes long recordings consistently.

Project Objectives and Scope

Objectives

- Process and segment ECG signals into individual beats.
- Extract timing, wavelet and statistical features.
- Train and compare four machine learning classifiers.
- Evaluate performance using class-sensitive metrics.

Project scope

- Dataset: MIT-BIH Arrhythmia Database.
- Four beat classes: N, S, V and F.
- Record-separated training and testing.
- Comparison of Logistic Regression, Random Forest, XGBoost and SVM.

Existing Challenges and Proposed Solution

Challenge	Proposed approach
ECG signals contain noise and variation	Band-pass filtering and beat normalization
Individual beats have complex patterns	RR timing, wavelet and statistical feature extraction
Normal beats dominate the dataset	Random undersampling and SMOTE inside the training pipeline
Different classifiers perform differently	Compare four ML models using multiple metrics
Familiar-record performance may mislead	Evaluate on a separate set of unseen records

Proposed pipeline

1 ECG data › 2 Preprocess › 3 Extract features › 4 Train models › 5 Classify beats

Dataset Overview

MIT-BIH Arrhythmia Database: a public ECG dataset for studying cardiac rhythm and heartbeat classification.

48	360 Hz	2	MLII
half-hour recordings	sampling frequency	channels per recording	primary lead used

Four target beat classes

Class	Description
N Normal	Normal and selected normal-like beats
S Supraventricular	Supraventricular ectopic beats
V Ventricular	Ventricular ectopic beats
F Fusion	Fusion of ventricular and normal beats

Experimental split

DS1 · Training

22 records · 51,002 usable beats

DS2 · Testing

22 records · 49,691 usable beats

Note: four paced records (102, 104, 107, 217) were excluded; training and testing use separate records.

Proposed System Architecture

1 ECG dataset

ECG recordings, MLII lead, beat annotations

2 Signal preprocessing

0.5–40 Hz filtering, beat segmentation, normalization

3 Feature engineering

35 RR timing, wavelet and statistical features

4 Training pipeline

Scaling, feature selection, class balancing

5 Model training

Logistic Regression, Random Forest, XGBoost, SVM

6 Evaluation and prediction

N / S / V / F, accuracy, macro-F1, MCC

End-to-end flow from raw recording to predicted beat class. Training and validation use record-grouped splits, so no patient appears on both sides.

ECG Preprocessing and Beat Segmentation

1 Input signal

Load the ECG recording and the annotated R-peak locations.

2 Noise filtering

Third-order Butterworth band-pass, 0.5 to 40 Hz, applied with zero-phase filtering.

3 Beat segmentation

Extract 90 samples before and after each annotated R-peak.

Key preprocessing parameters

0.5–40 Hz	180 samples	0.5 s	Per-beat z-score
filter range	window length	duration per beat	normalization

Takeaway: at 360 Hz, 180 samples represent about 0.5 s of ECG. Beats too close to a recording boundary are skipped when a full window cannot be extracted.

Converting ECG Signals into Meaningful Features

The model does not learn from the raw waveform. Each beat is described by 35 engineered features.

6

RR timing features

Intervals between heartbeats and their relative timing patterns.

17

Wavelet approximation

Daubechies-4 (db4) level-4 coefficients capture waveform structure.

4

Wavelet detail energies

Signal energy across selected detail-frequency bands.

8

Statistical features

Max, min, peak-to-peak, std, skewness, kurtosis and extremum positions.

Total features per beat: 6 + 17 + 4 + 8 = 35

Then StandardScaler scales them and SelectKBest (ANOVA F-test) keeps the most informative 15, 25 or 35.

Model Training and Evaluation Strategy

Training workflow

1. Scale numerical features.
2. Select features using the ANOVA F-test.
3. Apply RandomUnderSampler to reduce the majority class.
4. Apply SMOTE to enlarge minority classes.
5. Train and tune models with grouped cross-validation.

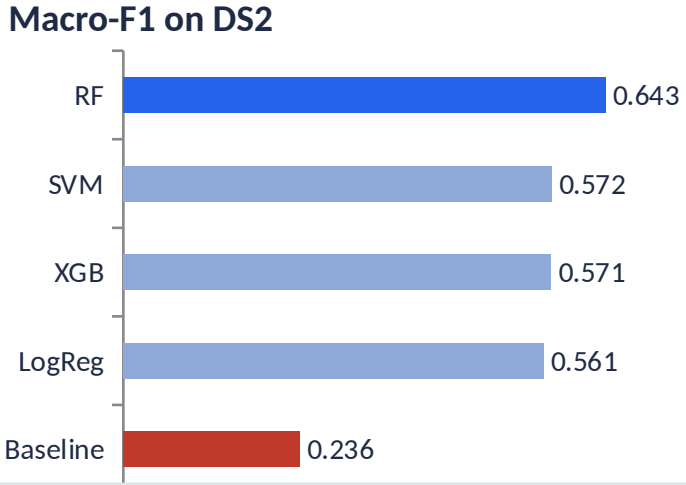
Models evaluated

Logistic Regression	Linear baseline
Random Forest	Ensemble of decision trees
XGBoost	Gradient-boosted trees
SVM (RBF kernel)	Nonlinear decision boundary

All steps are fitted inside training folds only, so nothing leaks from validation data. Tuning on DS1 with record-grouped folds; final scores on the separate DS2 test set.

Experimental Results: Model Comparison

Model	Accuracy	Balanced acc.	Macro-F1
Random Forest	**90.62%**	**0.7724**	**0.6431**
XGBoost	87.85%	0.6589	0.5707
SVM (RBF)	75.12%	0.6589	0.5715
Logistic Regression	78.06%	0.6281	0.5608
Majority baseline	89.04%	0.2500	0.2355

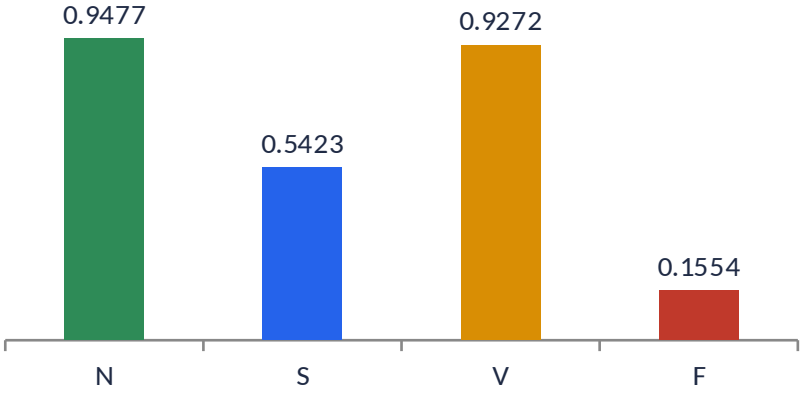


Key observations

- Random Forest gave the strongest DS2 performance, with macro-AUC 0.9587 and MCC 0.6531.
- The majority baseline reached 89.04% accuracy by favouring Normal beats, yet its macro-F1 was only 0.2355.
- Macro-F1 and balanced accuracy are more informative than accuracy on imbalanced classes.

Class-Wise Performance and Error Analysis

Random Forest: F1 per class



Confusion matrix (rows = true class)

	N	S	V	F
N	**40,476**	1,532	241	1,997
S	498	**1,272**	53	14
V	69	49	**3,080**	22
F	133	1	50	**204**

What do the results mean?

N: strong, F1 0.9477.

V: strong, F1 0.9272, recall 0.9565.

S: moderate, F1 0.5423; still hard to separate.

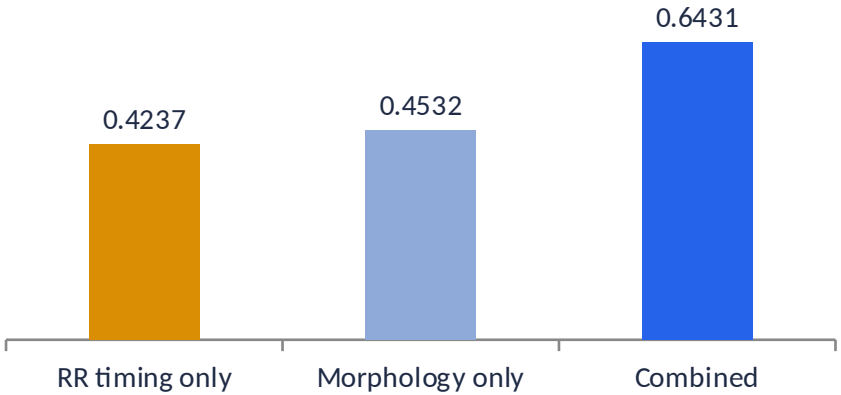
F: weak, F1 0.1554, precision 0.0912, so many false alarms.

Error analysis: record 233

1,499 normal beats were predicted as Fusion. This patient's beats arrive faster (mean RR about 0.59 s versus 0.85 s), so timing cues were misread. Excluding it lifts macro-F1 from 0.6431 to 0.6960. Fusion stays hard: only 414 training examples and a shape between N and V.

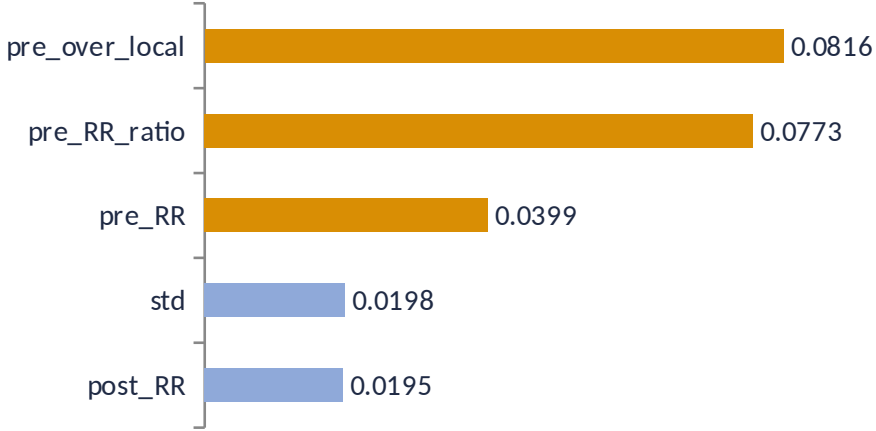
Feature Importance and Ablation Study

Which feature groups contributed? (macro-F1)



Accuracy: 82.15% · 82.66% · 90.62%

Most important features (permutation importance)



Amber = RR timing features

Main findings

- Timing finds early beats; shape separates ventricular beats. Neither group alone is enough.
- The top features describe timing relative to neighbouring beats.

Limitations and Future Scope

Current limitations

Class imbalance: rare classes, especially Fusion, remain hard to detect.

Record-specific errors: record 233 contributes heavily to misclassification.

Limited generalization: evaluation uses the MIT-BIH database only.

Real-time constraints: record-level RR normalization may use information unavailable live.

Clinical validation: a research prototype, not a validated diagnostic tool.

Future scope

- Explore 1D-CNN and CNN–LSTM models for automatic representation learning.
- Improve minority-class detection with advanced imbalance handling.
- Investigate multi-lead ECG and causal features for real-time inference.
- Validate on independent datasets such as INCART and MIT-BIH SVDB.
- Add calibrated probabilities and confidence-based alerts.

Conclusion and Key Findings

A machine learning pipeline classifies ECG heartbeats into Normal (N), Supraventricular (S), Ventricular (V) and Fusion (F) beats, and is wrapped in a working prototype app.



Prototype app (Random Forest) on record 200 of DS2. A single-record demo, not the overall test score.

Key findings

End to end: raw ECG to classified beats.

Complementary: timing and shape beat either alone.

Strong: Normal and Ventricular beats.

Weak: Supraventricular and Fusion beats need more data and features.

Thank you

Questions?

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